

AI Semester Project – Phase 1

Project Title: AI for Email Topic Categorization

Dataset Source (30 Emails)

For this project, a manually created dataset containing 30 emails was used. These emails were split into two categories:

Spam (15 emails): Messages that are promotional, fake offers, or scams.

Ham (15 emails): Normal emails such as work updates, meeting schedules, or personal messages.

This sample was inspired by public email datasets available on Kaggle and the UCI Machine Learning Repository. The dataset was made small and simple so that the algorithms could be implemented from scratch easily.

Libraries Used

Only the allowed libraries (as per the instructions) were used for:

pandas – to create and manage the dataset in table format.

numpy – for mathematical operations and array handling.

matplotlib – to create charts and visualize the performance of models.

CountVectorizer from sklearn – to convert text into numbers (Bag of Words). This is allowed because it's a preprocessing tool, not a machine learning model.

Algorithms Implemented (From Scratch)

The following machine learning algorithms were implemented manually, without using any built-in ML functions:

K-Nearest Neighbors (KNN) – used for classifying new emails based on similarity with training examples.

Decision Tree – a model that learns simple decision rules from the data to classify emails.

Linear Regression – applied on a separate dummy dataset to simulate regression analysis, as required in the project.

Evaluation Metrics

For Classification Models (KNN & Decision Tree):

Accuracy: Measures how many predictions were correct.

Precision: Measures how many of the predicted spam emails were actually spam.

Recall: Measures how many of the actual spam emails were correctly predicted.

For Regression Model (Linear Regression):

RMSE (Root Mean Squared Error): Tells how far the predictions are from the real values.

MAE (Mean Absolute Error): The average of the absolute differences between predictions and actual values.

R^2 Score (Optional): Shows how well the model explains the variation in the data.